

Vattanary Tevy

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SUMMARY

MS Mechanical Engineering candidate (UW, June 2026, Data Science & Mechatronics) with 3+ years of end-to-end ML and robot learning experience spanning real-hardware policy deployment, production-scale data analysis, and embodied AI research. Own a full robot learning pipeline at the UW WEIRD Lab; ROS2, LeRobot datasets, diffusion policy training in PyTorch/CUDA, and closed-loop evaluation on force-controlled UR5e arms with DAgger-style human-in-the-loop intervention. Spans data collection through hardware deployment and monitoring. Confirmed a month-long Amazon Robotics production failure in under 2 weeks using statistical root-cause analysis across millions of robotic cycles. Holds an Amazon Robotics Research Fellowship. Seeking a Research Engineer role to build and deploy learned robot policies and embodied AI systems.

EXPERIENCE

WEIRD Lab — Washington Embodied Intelligence & Robotics Development | Seattle, WA **Jan 2026 – Present**
Robot Learning Researcher

- Own config-driven robot learning pipeline (rosvbag → LeRobot dataset → training) and pytest infrastructure.
- Train diffusion policies in PyTorch/CUDA on teleoperated UR5e demonstrations for carbon fiber layup, ablating 1D U-Net and transformer backbones; deploy closed-loop at 10 Hz with an in-contact force loop fused from 6-DoF end-effector pose and wrist force/torque over RTDE.
- Evaluate rollouts against laser-profilometer scans of the workpiece and add DAgger-style human-in-the-loop intervention for failure-driven data collection.

Amazon Robotics | Seattle, WA **Sep 2025 – Dec 2025**
Robotics Data Scientist Intern

- Designed cross-model consistency metrics for segmentation, depth, and dimension pipelines on AWS S3 and SQL, enabling ML monitoring across millions of robotic cycles.
- Applied chi-square and independence tests to production logs to root-cause a failure mode unconfirmed for over a month, confirming it in under 2 weeks and unblocking the fix.

UW Sun Lab | Seattle, WA **Sep 2024 – Apr 2025**
ML Engineer & Researcher -- Amazon Robotics Research Fellow (University of Washington, 2024–2025)

- Deployed a real-time computer vision pipeline for automated concentration monitoring in a self-driving lab, replacing manual spot-checks with closed-loop feedback for thin-film deposition.

uWAMIT – Ultrasound-based WA Molecular Imaging & Therapy Center | Seattle, WA **Jun 2024 – Mar 2025**
Mechanical Design & Mechatronics Researcher

- Owned the mechanical design lifecycle of a photoacoustic-ultrasound system (GE HealthCare) and built a 1000 Hz multimodal acquisition system, resolving a laser-fiber synchronization instability with an embedded pulse-timing redesign.

TECHNICAL SKILLS

- **Robot Learning:** PyTorch (CUDA), diffusion policy, imitation learning (ACT, behavior cloning), DAgger, LeRobot, W&B sweeps
- **Robotics & Control:** ROS2, bimanual UR5e, URDF, FK/IK, impedance control, force/torque control (RTDE), laser profilometry
- **Software & Mechanical:** Python, SQL, AWS S3, statistical hypothesis testing, C++, pytest, Linux, MATLAB, SolidWorks, GD&T, FEA, DAQ

EDUCATION

University of Washington | Seattle, WA **Jun 2026**
M.S. in Mechanical Engineering – Data Science & Mechatronics